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A comparative evaluation of multiple machine learning approaches for forecasting dengue outbreaks in Bangladesh

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This study aims to forecast dengue incidence in Bangladesh by applying and comparing machine learning techniques. Dengue surveillance data from January 1, 2022, to December 1, 2023, for five divisions of Bangladesh was obtained from the Directorate General of Health Services. Initial time series analysis showed distinct trends and seasonality. Seasonal Autoregressive Integrated Moving Average (SARIMA), Multi-Layer Perceptron neural networks, XGBoost, and Support Vector Regression (SVR) were implemented for modeling and forecasting monthly dengue cases for each division. The models were evaluated using error metrics like RMSE, MAE, and MAPE. The results indicate that XGBoost provided the most accurate forecasts with the lowest errors overall. For the Dhaka division, which had the most data, XGBoost captured seasonality and trends well, with an RMSE of 109, an MAE of 127, and an MAPE of 12.9%. SARIMA models were reasonably good but had higher errors than the machine learning models. SVR was found unsuitable for forecasting this data. In summary, advanced machine learning models like XGBoost proved effective for infectious disease forecasting using limited surveillance data in resource-constrained settings. The forecasts can assist public health planning in Bangladesh.

Keywords Dengue, Forecasting, Machine learning, SARIMA, Neural networks, XGBoost, SVR

A type of mosquito-borne disease known as dengue fever (DF) or dengue hemorrhagic fever (DHF) causes a high-temperature, sudden illness[1]. Serotypes 1–4 of the Aedes mosquito are responsible for the more dangerous dengue fever[2]. Dengue fever, which is spread by mosquitoes, is a major public health concern globally. Half of the global population and 129 nations are at risk of illness[3]. Every year, somewhere around 100–400 million illnesses are recorded worldwide[4]. Dengue fever symptoms are similar to the flu and can be moderate to severe. Because there are four different serotypes of dengue, an individual may get up to four infections throughout their lifetime, and a severe case of the disease can be fatal. There is currently no universal vaccination or targeted treatment; so, vector control remains the primary means of prevention. The importance of disease surveillance is highlighted by this. Public health resources can be optimally allocated with the use of prediction algorithms to combat dengue and reduce overall sickness burden[5]. The incidence of dengue fever is associated with many risk factors. These include meteorological variables such as temperature, precipitation, and severe weather events[6–8]. The literature on dengue forecasting integrates multidisciplinary information from epidemiology, environmental science, computer science, and mathematics. Modeling frameworks include both empirical and theoretical approaches. Compartmental models utilize extensive data to estimate the temporal dynamics of dengue infections within a population, elucidating the transmission pathway, the vector, and the host. Agent-based modelling, a simulation technique, is effective for assessing the impact of actions such as the release of sterile males to diminish the mosquito population[9]. A data-centric approach that captures the highly autocorrelated nature of dengue illness is statistical time series forecasting[10,11].

Emerging as a worldwide public health issue is dengue, the fastest-growing mosquito-borne infectious illness[12]. One of Bangladesh's biggest public health issues is dengue, which is endemic and often breaks out. The dengue virus has the capacity to start epidemics that have high rates of morbidity and fatality. Sanofi Pasteur's

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dengue vaccine has been approved in 24 countries and included in public immunization programs in Brazil and the Philippines, so this is especially worrisome because there is now no widely available therapy or vaccine against dengue[13]. According to the World Health Organization, dengue fever is one of the top 10 most dangerous dangers to global health[14]. Prior to 1970, dengue was a concern in just nine countries; today, it is present in over 100 nations across WHO regions, with 2016 being recognized as a major dengue outbreak year globally[15].

Since the first recorded epidemic in 2000, the South Asian country of Bangladesh, home to more than 165 million people, has been plagued by dengue. Multiple risk factors have contributed to the current string of devastating dengue epidemics in Bangladesh. Public health catastrophes like dengue fever could happen if we don't do something about our outdated healthcare system, our lack of readiness for outbreaks, and the general public's ignorance of the disease.

It is anticipated that dengue fever may spread farther into countries with people that are immunologically naive, such as sub-Saharan Africa, sections of Europe and the northern United States of America, and lowland areas of the Western Pacific and Eastern Mediterranean regions, among other places, as a result of the changing climate brought on by global warming[16]. There will be approximately 1.4 billion more individuals at risk, bringing the total number of people who will be at risk to 4.7 billion. Additionally, it is anticipated that the climate appropriateness for dengue transmission will rise by four extra months[17].

The current situation in 2020 DF also attacks as many as 3 million people in the South Asia region, but with determination, by making thought of earlier precaution to reduce the number of diseases[18].

This study analyzes the prediction accuracy and computational efficiency of sophisticated machine learning models compared to more traditional models on dengue datasets and to find out which characteristics (such as weather, humidity, and rainfall) have the biggest impact on dengue incidence with feature selection techniques.

Literature review

In forecasting dengue infection, relative humidity is unnecessary, and climatic parameters have more rapid and long-term effects than short-term ones[40]. Various methodologies for predicting dengue fever have been explored in the research domain, utilizing machine learning predictions based on certain traceable behavioural reactions. Numerous countries have conducted analyses on machine learning in relation to dengue cases in recent years. Martheswaran et al.[19] used case report data from 2012 to 2020 to estimate dengue incidence in Singapore and Honduras utilizing the random-sampling-based susceptible-infected-removed (SIR) model. The proposed model was calibrated utilizing the Bayesian Markov Chain Monte Carlo (MCMC) method. The study's findings indicated that its strategy may be applied to other outbreaks, including the current COVID-19 pandemic, to enhance understanding of the outbreak's future dynamics. The difficulties of data discontinuity and paucity in epidemiological and healthcare datasets are examined by Mobin et al.[42]. Based on the underlying distribution, the study uses a new data downscaling algorithm called MahadevKamrujjaman Downscaling (MKD), also known as Stochastic Bayesian Downscaling (SBD).

Al-Mobin[20] pioneered stochastic Bayesian down-scaling (SBD) to convert 168 monthly records into 5113 daily points, proving that random forests on down-scaled data can reach 95.8% accuracy 28.5 pp higher than on raw data when only meteorological covariates are used. Nonetheless, SBD was applied to national-level counts and did not evaluate model transferability across sub-regions or compare classical and deep networks. The revised manuscript extends this contribution by testing SARIMA, MLP, and XGBoost on divisional series with heterogeneous dynamics, thereby examining model robustness beyond national aggregates.

Rahman et al.[21] integrated climate, socio-demographic and landscape indicators over 2000–2021, ran Light, XGBoost and RF with tenfold CV, and used SHAP to expose nonlinear thresholds (e.g., T_min 22 °C; RH 82%) for Bangladeshi dengue risk. The study, however, delivered national forecasts only, left daily granularity and cross-division heterogeneity unexplored, and did not benchmark against classical time-series baselines. Our revised manuscript plugs these holes by (i) producing division-specific predictions, (ii) contrasting tree ensembles with SARIMA and MLP, and (iii) quantifying performance via MAE/RMSE/MAPE for each division.

Using daily DGHS/IEDCR surveillance, Hossain et al.[22] quantified 321,179 cases and 1705 deaths, revealed sex-specific mortality differences, and confirmed Dhaka's disproportionate burden. While the paper is indispensable for descriptive epidemiology, it stopped short of predictive modeling or variable interaction analysis. The revised manuscript leverages this rich spatial–temporal case archive as training data and couples it with climatic drivers to move from description to forecast, thereby filling the predictive analytics gap.

Al-Mobin[23] devised a dynamic pipeline that combined 13 meteorological predictors with six machine-learning algorithms and introduced the Sequential Squeeze Feature Selection (SSFS) wrapper, boosting decision-tree accuracy to 84.02% and reducing MAPE by 70.82%. Although the study advanced feature-engineering practice and showed that relative humidity was redundant, it remained confined to monthly data, a single-factor (meteorology) focus, and did not test ensemble models or fine-grained spatial variation. This research addresses these gaps by comparing multiple learning paradigms (SARIMA, XGBoost, MLP) on division-level series, benchmarking against ensemble results and adding socio-demographic covariates.

Predicting dengue cases in Bangladesh using Support Vector Machine (SVM) and Random Forest (RF) was the goal of a machine learning-based comparative study developed by Dourjoy et al.[24]. For this study, the authors used online survey data from 600 patients. The results showed that RF achieved 64% accuracy and SVM 68% performance. Although other elements such as metrological data, socioeconomic effect, geographic location, and weather correlation with monthly instances were taken into consideration, their study simply focused on the pattern of the patient's data. Additionally, these findings have not been validated.

Uno et al.[25] performed a study in the United States to examine the impact of the dengue virus on the human body. This research illustrates the life cycle of the dengue virus within the human body, revealing that dengue vaccination can diminish the incidence of dengue cases. Conversely, the dengue vaccine has faced significant criticism for its limited efficacy in disease prevention, achieving only 60% effectiveness.

Sunitha et al.[26] have delineated the mining model for dengue fever. This paper implements KNN. This paper enhances mining performance. They have utilized the model to segment microscopic blood images, employing a neural network to enhance outcomes related to dengue illness. Application of back propagation networks to categorization outcomes. It provides 98% accuracy.

Althouse et al.[27]applied linear regression and two classifications of logistic regression and SVM to forecast over-surpassing dengue incidence in the Bangkok and Singapore regions. Outlook for dengue datasets reaches the internet data source for surveillance system prediction. Almost five years of data produced accuracy based on dengue events anticipated by the binary categorization. Classification analysis helps LR and SVR to be compared between R^2 value and correlation fitted value.

Aburas et al.[28]conducted an examination of dengue prediction on confirmed cases with an ANN model estimation. In Malaysia, the region impacted by the disease has been extensive from January 2001 to 2007. There is concern regarding dengue cases reported by the NEA, which is publicly available from Malaysia, indicating approximately 14,209 instances correlated with ANN, exhibiting a strong correlation coefficient of 0.96. Neural networks necessitate training on input sequences and a test dataset divided into 70% for training and 30% for testing. Hidden neurons are determined by four features, such as temperature and humidity, which generate the correlation coefficient and RMSE error. Wind speed was negatively linked to dengue fever cases within the same month, whereas sunshine hours were either correlated[43] or inversely correlated with dengue fever incidence.

Linear regression, a machine learning approach, was announced by Karim et al.[29] to describe the situation in Dhaka city, Bangladesh, from 2000 to 2008. Evidence from model validation suggests that climate change is a major contributor to the dengue epidemic in Dhaka.

Using multiple linear regression, Karim et al.[41] created a dengue detection model that was retrospectively validated using data from 2001, 2003, 2005, and 2008. Monthly data from the Directorate General of Health Services (DGHS) and the Meteorological Department of Dhaka, Bangladesh, covering the years 2000–2008, were used in the study.

Research on dengue prediction rarely uses advanced deep learning models like MLP and XG-Boost, which excel at handling complex time-series data, favoring traditional machine learning methods such as random forests and SVM instead. A detailed comparison of these approaches could uncover improvements in prediction accuracy. Exploring these differences may lead to a more effective and tailored strategy for dengue prediction in Bangladesh, enhancing public health efforts. Furthermore, we assess the effectiveness of traditional time series models versus machine learning techniques, a comparison often neglected in previous studies. This approach offers a fresh perspective for forecasting dengue cases in the Bangladeshi population.

Methodology

Data

Data is sourced from a secondary repository, the Directorate General of Health Services (DGHS) of Bangladesh, and gathers data from the dengue press release on a daily basis. This research aims to analyze dengue data from January 2022 to December 2023. The data is collected across all five divisions of Bangladesh. Data is obtained from the website: https://old.dghs.gov.bd/index.php/bd/

Training and testing datasets: It is expected that the test data will give a trustworthy indication of how well the model is likely to forecast on new data, since it is not used to determine the forecasts. Although this value varies depending on the length of the sample and the forecasting horizon, the test set size is normally approximately 30% of the total sample. Ideally, the size of the test set should match the maximum forecast horizon that is necessary[30]. Thus, first 70% of our data set is training data and remaining 30% is the test data set.

Feature selection criteria: The process of feature selection holds great significance in the realm of ML model development. It allows the identification of crucial variables from extensive datasets that have a significant influence on the model in comparison to other variables. In our study, we utilized scores to identify these important variables such as temperature, rainfall, and humidity.

Methods

Seasonal ARIMA (SARIMA)

This section outlines the fundamental principles of the Autoregressive Integrated Moving Average (ARIMA) model. The ARIMA model, denoted as $(\mathrm{p}, \mathrm{d}, \mathrm{q})$, is characterized by three components: d represents the degree of differencing, p indicates the order of the autoregressive part, and q signifies the order of the moving average component[31]. The formulation of the ARIMA model is presented in Eq. (1) as

$$z_t = \delta + \phi_1 z_{t-1} + \phi_2 z_{t-2} + \cdots + \phi_p z_{t-p} + a_t - \theta_1 a_{t-1} - \theta_2 a_{t-2} \cdots - \theta_q a_{t-q} \tag{1}$$

where z_t is level of differencing, the constant is notated by δ, while ϕ is an autoregressive operator, a is a random shock corresponding to time period t, and θ is a moving average operator.

Seasonal ARIMA (SARIMA) is applicable when a time series displays seasonal fluctuations. The model incorporates both a seasonal autoregressive component (P) and a seasonal moving average component (Q) resulting in the multiplicative structure of SARIMA represented as $(p, d, q)(P, D, Q)_s$.The subscript 's' indicates the duration of the seasonal cycle. For instance, for hourly data, is equals 7; for quarterly data, s equals 4; and for monthly data, s equals 12. To formalize the model, the backshift operator (B) is employed. The notation B^k,denotes the observation of the time series shifted back by k periods, expressed as $B^k y_t = y_{t-k}$. Historically, the backshift operator has been utilized to illustrate a general transformation towards stationarity, where a time series is considered stationary if its statistical characteristics, such as mean and variance, remain constant over time. The general transformation for stationarity is outlined below.

$$z_t = \nabla_s^D \nabla^d y_t = (1 - B^s)^D (1 - B^s)^d y_t \tag{2}$$

where z is the time series differencing, d is the degree of nonseasonal differencing used, and D is the degree of seasonal differencing used. Then, the general form of SARIMA model SARIMA $(\mathrm{p}, \mathrm{P}, \mathrm{q}, \mathrm{Q})$ is,

$$\phi_p\left(B\right)\phi_p\left(B^s\right)z_t = \delta + \theta_q(B)\theta_Q(B^s)a^t \tag{3}$$

George Box and Gwilym Jenkins conducted an analysis of streamlined procedures aimed at acquiring a thorough comprehension of the ARIMA model, particularly focusing on the application of the univariate ARIMA model[31,32]. The Box-Jenkins (BJ) methodology is characterized by a sequence of four iterative steps:

1. Step 1: Identification.
2. Step 2: Estimation
3. Step 3: Diagnostic checking
4. Step 4: Forecasting

Extreme gradient boosting (XGBoost)
Extreme gradient boosting (XGBoost) is an ensemble ML technique that uses decision trees to provide a gradient boosting framework called Extreme Gradient Boosting (XGBoost). To reach the final prediction, XGBoost creates new models that predict the residuals of the earlier models[33].

Support vector regression (SVR)
Support vector regression (SVR) is an ML algorithm that uses supervised learning models to solve complex classification, regression, and outlier detection problems by performing optimal data transformations that determine boundaries between data points based on predefined classes, labels, or outputs. SVRs are widely adopted across disciplines such as healthcare, natural language processing, signal processing applications, and speech & image recognition field[34]. The equation for the linear hyperplane can be written as:

$$w^t x + b = 0$$

where w is the normal vector to the hyperplane (the direction perpendicular to it). b is the offset or bias term, representing the distance of the hyperplane from the origin along the normal vector.

Multi-layered perceptron (MLP)
Multi-Layered Perceptron (MLP) is a supplement to the feed forward neural network. It consists of three types of layers—the input layer, the output layer, and the hidden layer. The input layer receives the input signal to be processed. The required tasks, such as prediction and classification, are performed by the output layer. An arbitrary number of hidden layers that are placed in between the input and output layer are the true computational engine of the MLP. Similar to a feed forward network in an MLP, the data flows in the forward direction from the input to the output layer. The neurons in the MLP are trained with the back propagation learning algorithm. MLPs are designed to approximate any continuous function and can solve problems that are not linearly separable. The major use cases of MLP are pattern classification, recognition, prediction, and approximation. The multilayer perceptron (MLP) is used for a variety of tasks, such as stock analysis, image identification, spam detection, and election voting predictions[35].

Modeling approach

- Training/Test split: 80/20
- Specification of Hyperparameters:

All the hyper parameters are selected based on the Grid Search Cross Validation approach.

1. Random Forest:

   n_estimators = 100, random_state = 42

2. XGBoost:

   base_score = 0.5, booster='gbtree', n_estimators = 1000,early_stopping_rounds = 50,objective='reg:linear',learning_rate = 0.1

3. SVR:

   C=1000, gamma = 0.01,kernel='rbf'

4. MLP:

   hidden_layer_sizes = (2,2),max_iter = 5000,activation='relu',solver='adam',random_state = 42

Performance metrics

Augmented Dickey-Fuller test

The Augmented Dickey-Fuller (ADF) test enhances the original Dickey-Fuller test by including lagged differences of the time series, effectively tackling the problem of higher-order autocorrelation[36]. The primary aim of the ADF test is to assess the null hypothesis that a unit root is present in the time series, suggesting that the series is non-stationary and exhibits a trend. This null hypothesis is evaluated using t-statistics, calculated according to the following formula:

$$t = \frac{\widehat{\gamma} - \gamma}{SE\left(\widehat{\gamma}\right)}$$

When the computed t-value surpasses the critical threshold, the null hypothesis remains accepted. In this case, the variable is identified as non-stationary, exhibiting a unit root. In contrast, if the computed t-value is below the critical threshold, the null hypothesis is rejected, signifying that the series is stationary and devoid of a unit root. The initial evaluation of the series occurs at its level; should it not display stationarity, additional tests are performed on the first and second differences in succession. An alternative method for deciding on the rejection of the null hypothesis involves analyzing the calculated value's position in relation to the critical value. In a one-tailed test, if the calculated value is positioned to the right of the critical value, the null hypothesis is upheld; conversely, if it is to the left, the null hypothesis is rejected, indicating the absence of a unit root in the series. Furthermore, the p-value acts as a determinant in hypothesis testing: a p-value lower than 0.05 results in the null hypothesis being rejected, whereas a p-value of 0.05 or higher leads to its acceptance.

The Akaike information criterion (AIC)

The Akaike Information Criterion (AIC) is a key instrument for selecting models in statistics. It assesses the comparative quality of different statistical models[37]. Essentially, AIC helps to pinpoint the model that best explains the data while employing the fewest parameters. The formula for AIC is:

$$AIC = 2k - 2ln(L)$$

where k is the number of parameters in the model and L is the likelihood of the model given the data.

Mean absolute error (MAE)

The MAE score is measured as the average of the absolute error values. Therefore, the difference between an expected value and a predicted value can be positive or negative and will necessarily be positive when calculating the MAE[38]. This metric can be written as:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|Y_i - \widehat{Y}_i\right|$$

Root mean square error (RMSE)

The root-mean-square deviation (RMSD) or root-mean-square error (RMSE) serves as a commonly used metric for assessing the alignment between projected or estimated values. The RMSE of an estimator defined as,

$$\mathrm{RMSE} = \sqrt{\mathrm{MSE}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2}$$

Values of MSE may be used for comparative purposes. RMSE is a good measure of accuracy for evaluating the quality of predictions and shows how far predictions fall from measured true values using Euclidean distance[39].

Mean Absolute Percentage Error (MAPE)

The mean absolute percentage error (MAPE) is a statistical measure of how accurate a forecast system is. It measures this accuracy as a percentage and can be calculated as the average absolute percent error for each time period minus actual values divided by actual values. Where A_t is the actual value and F_t is the forecast value, this is given by:

$$MAPE = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{A_t - F_t}{A_t}\right| \times 100$$

The mean absolute percentage error (MAPE) is the most common measure used to forecast error and works best if there are no extremes to the data[40].

Software

We use of R version 4.3.3 for SARIMA modeling and Python 3.13 for machine learning tasks, such as implementing MLP. Importantly, they point out that MLP outcomes may vary depending on the CPU architecture in use. As a result, they emphasize the importance of detailing hardware specifications to support accurate replication of results. Such transparency is essential for research reproducibility and enables meaningful comparisons across diverse computational settings.

Justification for SARIMA over SARIMAX

We purposefully used SARIMA rather than SARIMAX as the statistical benchmark, even though exogenous factors like temperature, humidity, and rainfall were accessible and subsequently included in the machine learning models. Creating a totally autoregressive baseline based only on past dengue incidence was the main driving force. In contrast to more sophisticated models that explicitly incorporate external factors, this enabled us to establish a clear and understandable point of comparison.

Furthermore, although our dataset's exogenous variables were instructive, they weren't always precisely aligned across divisional dimensions and temporal lags. They ran the danger of producing misleading correlations and unstable parameter estimates when incorporated into SARIMAX. Because of this, we limited the statistical benchmark to SARIMA, which is frequently used in epidemiological forecasting and has demonstrated efficacy in detecting autoregressive dynamics and seasonal cycles.

However, we recognize that SARIMAX is still a useful addition. Once higher-resolution and more consistently lagged exogenous data are available, further research should look into SARIMAX models since they could improve the precision of conventional time series forecasts even more.

Workflow of this study

Figure 1 illustrates this study's workflow. Initially, we train our dataset to identify significant features such as temperature, rainfall, and humidity. Subsequently, we implement a conventional time series model known as SARIMA. Following this, we apply two machine learning models, XGBoost and SVR, along with a deep learning model referred to as MLP, to our trained data. Finally, we generate forecasts from our models and assess their performance using metrics including MAE, RMSE, and MAPE.

Results and analysis

Feature selections

The following graph determines the most influential environmental features affecting dengue incidence, using feature selection and importance ranking methods within machine learning models. This research examines time series data, focusing on key variables that influence dengue incidence, including temperature, rainfall, humidity, and the number of patients admitted to the hospital.

To ensure robust and leakage-free feature selection for our time series data, we applied Recursive Feature Elimination with Cross-Validation (RFECV) using a Random Forest regressor as the estimator. Unlike methods

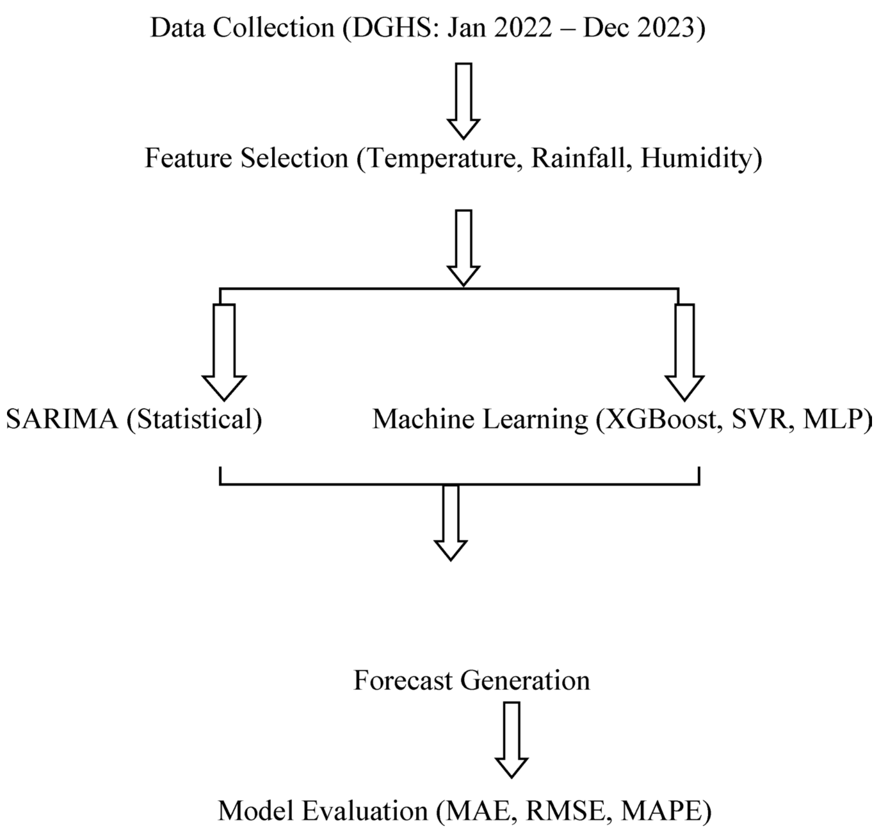


Fig. 1. Workflow of this study.

Metric of evaluation	Value
MAE	702.10
RMSE	755.74

Table 1. Performance of feature selection model.

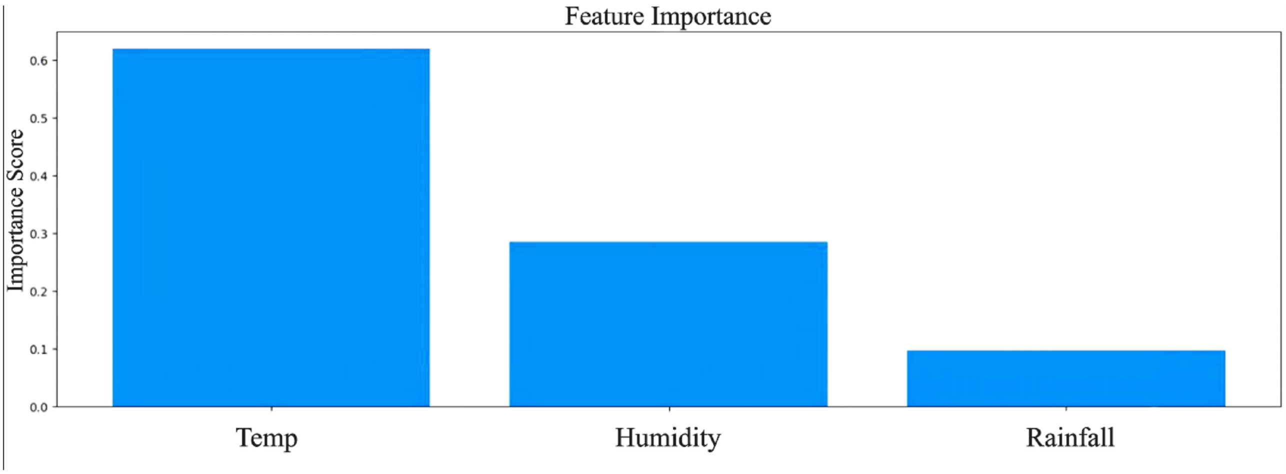


Fig. 2. Feature importance for dengue incidence prediction using Recursive Feature Elimination with Cross-Validation (RFECV).

Feature	Importance
Temperature	0.6
Humidity	0.3
Rainfall	0.1

Table 2. Performance of feature selection model.

such as Boruta, which may introduce data leakage by shuffling feature values and thus violating temporal dependencies, RFECV evaluates feature subsets based on predictive performance while preserving the chronological order of observations. We employed a time series-aware cross-validation strategy (TimeSeriesSplit) to ensure that model training always preceded validation temporally, thereby mimicking real-world forecasting scenarios. Model performance was assessed on a holdout set using mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination (R^2), and the results were visualized through actual versus predicted plots and feature importance rankings. This methodology follows best practices for time series machine learning as outlined by Hyndman and Athanasopoulos[41].

The dataset was analyzed to identify key features: temperature, humidity, and rainfall. Table 1 indicate that model's performance was evaluated, yielding an RMSE of 755.74, indicating a significant average discrepancy from actual values, yet robust compared to other models and the MAEvalue was 702.10.

This distribution, which is displayed in Fig. 2, indicates that temperature is likely the predominant element affecting the model's predictions, followed by humidity, although rainfall exerts minimal influence. This information may be advantageous for prioritizing data gathering or feature selection in subsequent model construction. With an important value of about 0.6 (Table 2), temperature (Temp) is the most important factor, indicating it performs the most significant influence in the predictions of the model.

Results of time series analysis using SARIMA model

Figure 3 shows time series graphs of reported cases in five divisions of Bangladesh such as Dhaka, Chattogram, Barisal, Khulna, and Other Divisions from January 2022 to November 2023. The graphs in Fig. 3a,b reveal distinct patterns, with Dhaka and Chattogram experiencing the highest peaks, likely due to seasonal outbreaks. From Fig. 3c–e,Barisal, Khulna, and Other Divisions show similar but less pronounced trends, indicating regional differences in outbreak intensity and timing. This data underscores the need for targeted public health strategies in the most affected areas. Additionally, Fig. 3 includes SARIMA model predictions for dengue cases in Dhaka, Chattogram, Barisal, and Khulna, based on time series data. The model's forecasts were compared to actual historical data, showing good alignment in Dhaka when supplemented with additional data, while Khulna's predictions were less consistent due to limited data. Overall, the time series data exhibited clear seasonal patterns

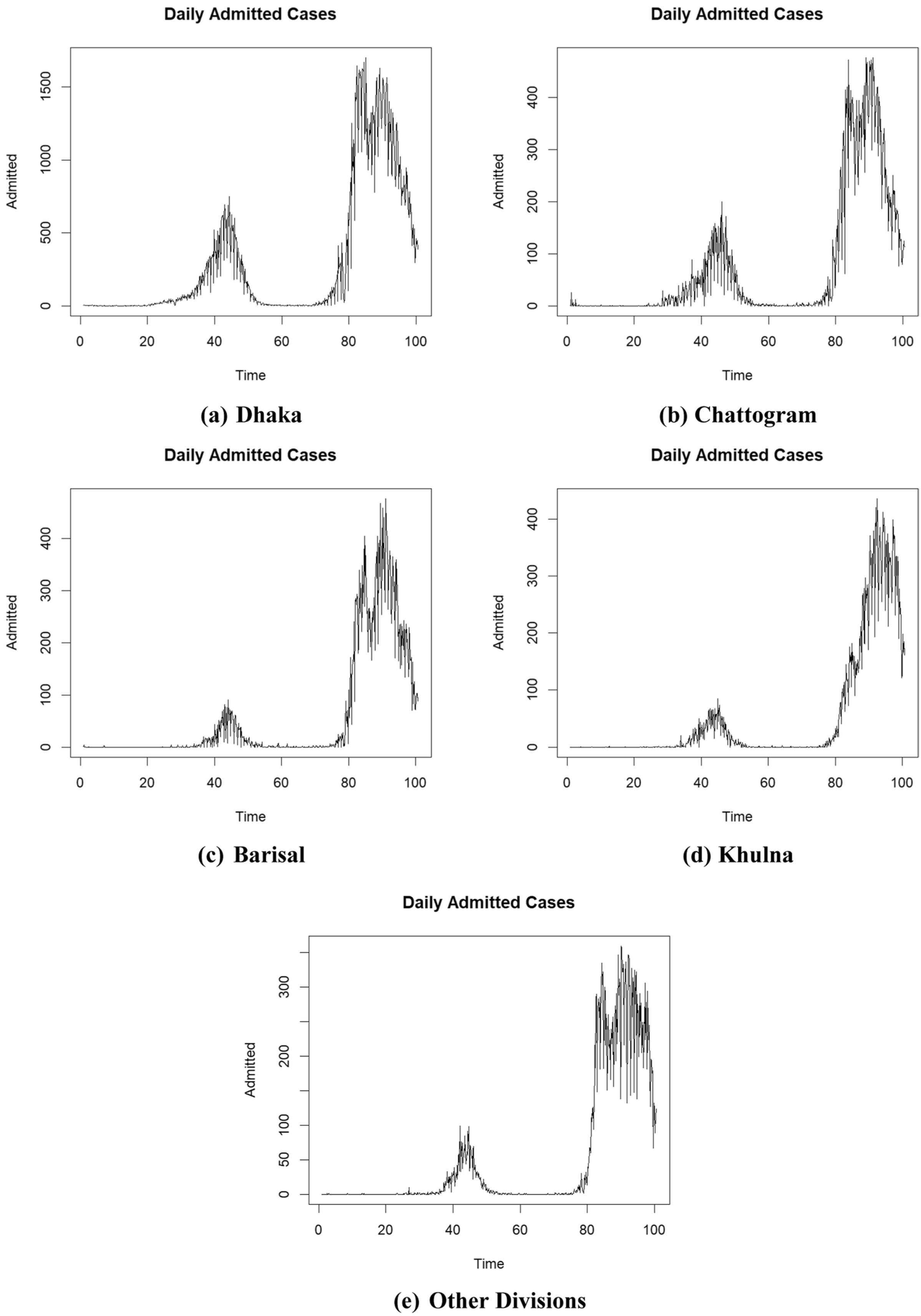


Fig. 3. Time Series Plot of No. of Cases.

Divisions	Model	AIC	Log Likelihood	RMSE	MAE	MAPE
Dhaka	$SARIMA(1{,}0,1)(1{,}1,1)$	7964	– 3977	75.864	39.518	0.545
Chattogram	$\mathrm{SARIMA}(0{,}1,3)(1{,}0,1)$	6545	– 3266	26.145	13.151	0.814
Barisal	$\mathrm{SARIMA}(1{,}0,1)(1{,}1,2)$	6212	– 3100	21.132	9.894	0.638
Khulna	SARIMA(4,0,0)(0,1,1)	5757	– 2872	15.241	7.158	0.500
Other divisions	$\mathrm{SARIMA}(1{,}0,1)(1{,}1,1)$	6128	– 3059	20.099	8.602	0.743

Table 3. Best model statistics using SARIMA model.

Divisions	Model	Coefficients
Dhaka	SARIMA(1,0,1)(1,1,1)	ar1 = 0.65, ma1 = – 0.60, sar1 = 0.85, sma1 = – 0.80
Chattogram	SARIMA(0,1,3)(1,0,1)	ma1 = – 0.40, ma2 = – 0.25, ma3 = – 0.18, sar1 = 0.85, sma1 = – 0.75
Barisal	SARIMA(1,0,1)(1,1,2)	ar1 = 0.9995, ma1 = – 0.6297, sar1 = 0.2283, sma1 = – 1.4463, sma2 = 0.4476
Khulna	SARIMA(4,0,0)(0,1,1)	ar1 = 0.82, ar2 = – 0.34, ar3 = 0.33, ar4 = – 0.22, sma1 = – 0.65
Other Divisions	SARIMA(1,0,1)(1,1,1)	ar1 = 1.00, ma1 = – 0.87, sar12 = – 0.03, sma12 = – 1.00

Table 4. The coefficient of the best performing SARIMA models.

that the SARIMA model effectively captured, and the Augmented Dickey-Fuller Test confirmed the stationarity of the variables used.

The Akaike Information Criterion (AIC) is a widely recognized statistical tool employed for the purpose of model selection. It facilitates the comparison of various candidate models to identify the most suitable one. Initially, models were developed for each distinct number of cases across all divisions in Bangladesh, resulting in the calculation of AIC values. The optimal model was determined based on these values, with a lower AIC indicating a superior model. This information is presented in Table 2.

Table 3 shows that the $SARIMA(1{,}0,1)(1{,}1,1)$ model for the Dhaka division has the highest AIC of 7964and the lowest log-likelihood of – 3977compared to other divisions. It also records the highest RMSE and MAE at 75.864and 39.518, respectively. The MAPE indicates an average deviation of 54.5% from actual values, the second lowest among the models. Conversely, for the Chittagong division, the SARIMA(0,1,3)(1,0,1) model presents a comparatively elevated AIC value of 6545and a lower log-likelihood value of – 3266among its counterparts. This model demonstrates lower RMSE and MAE values of 26.145and 13.151, respectively. According to MAPE, the forecasted values for the Chattogram division are, on average, 81.4% off from the actual values, which are regarded as the highest, among others. In the case of the Barisal division, the SARIMA(1,0,1)(1,1,2)model demonstrates a comparatively lower AIC value of 6212and a higher log-likelihood value of – 3100when evaluated against alternative models. This model also exhibits reduced RMSE and MAE values of 21.132and 9.894, respectively. The MAPE indicates that, on average, the forecasted values for the Barisal division deviate by 64% from the actual values, marking it as the second lowest value. The SARIMA(4,0,0)(0,1,1)model is the best for the Khulna division, with the lowest AIC of 5757and highest log-likelihood of – 2872, along with RMSE and MAE values of 15.241and 7.178. However, the MAPE shows an average 63.4% inaccuracy in predictions for Khulna, indicating poorer performance compared to other divisions. For the other divisions, the $SARIMA(0{,}1,1)$ model has a favorable AIC of 8129.29 and log-likelihood of 4061.64, but a higher RMSE of 82.266 and MAE of 17.277. The Mean Absolute Percentage Error (MAPE) shows an average discrepancy of 50% among divisions, indicating significant variability. While SARIMA models fit reasonably well according to the Akaike Information Criterion (AIC) and log-likelihood, they exhibit notable variability in forecast errors. The models for Dhaka are the least accurate based on Root Mean Square Error (RMSE) and Mean Absolute Error (MAE). Adding more data may enhance these models' performance.

From Table 4, the SARIMA model coefficients for dengue incidence across Bangladeshi divisions underscore distinct regional variations in temporal patterns. In the case of Dhaka, moderate autoregressive (ar1 = 0.65) and moving average terms (ma1 = – 0.60), together with pronounced seasonal effects (sar1 = 0.85, sma1 = – 0.80), indicate sustained incidence levels and a clear annual cycle.

Chattogram presents a notably intricate situation, with three distinct moving average components (ma1 = – 0.40, ma2 = – 0.25, ma3 = – 0.18) interacting alongside pronounced seasonality (sar1 = 0.85, sma1 = – 0.75). This combination points to substantial short-term variability layered over a stable seasonal pattern, highlighting the region's complex transmission dynamics.

Barisal, in contrast, is marked by an autoregressive coefficient almost equal to one (ar1 = 0.9995), which suggests an exceptionally persistent pattern of dengue transmission. The seasonal moving average terms (sma1 = – 1.45, sma2 = 0.45) further indicate nuanced fluctuations aligned with seasonal changes.

Turning to Khulna, the alternation in sign among autoregressive parameters (ar1 = 0.82, ar2 = – 0.34, ar3 = 0.33, ar4 = – 0.22), together with a notable seasonal moving average effect (sma1 = – 0.65), points to oscillatory behaviour in dengue incidence. Altogether, these findings make clear the substantial heterogeneity in transmission dynamics seen across different regions.

Other divisions tend to manifest a simpler configuration, characterized by a strong first-order autoregressive effect (ar1 = 1.00), a pronounced moving average component (ma1 = – 0.87), and distinct annual seasonality (sar12 = – 0.03, sma12 = – 1.00). This points to pronounced autocorrelation and a regular yearly cycle. Collectively, these coefficient patterns demonstrate that dengue transmission and its predictability vary considerably by region, which has important implications for modeling strategies and forecasting accuracy within each division.

Now, we use the selected model to forecast the dengue cases.

The forecasted graphs in Fig. 4 indicate a gradual downward trend across all divisions. The confidence intervals appear to be relatively narrow, suggesting that the models fit adequately. Nevertheless, it is important to note that these forecasts may be subject to modification as more data becomes available.

Machine learning prediction for dengue cases

Figure 5 illustrates the development of an XGBoost model for predicting dengue cases across various regions in Bangladesh, including Dhaka, Chattogram, Barisal, and Khulna. This advanced gradient boosting machine learning technique analyzes time series data on dengue incidence. After training, the model forecasts dengue cases for these regions, with plots comparing predictions to actual historical data. In Dhaka from Fig. 5a, projections align closely with seasonal trends when additional data is included, while in Khulna from Fig. 3d, where data is limited, the model shows considerable variability. Overall, the time series data reveals distinct patterns and seasonal variations that the XGBoost model effectively captures.

From Fig. 6, Support Vector Regressions (SVRs) were utilized to predict the incidence of dengue across the five divisions, such as Dhaka, Chattogram, Barisal, Khulna, and Other Divisions. For each division, unique forecast plots have been created to depict both the historical data and the projected incidence of dengue.

From Fig. 7, Multilayer Patron (MLP) was utilized to predict the incidence of dengue across the five divisions, such as Dhaka, Chattogram, Barisal, Khulna, and Other Divisions. For each division, unique forecast plots have been created to depict both the historical data and the projected incidence of dengue.

Table 5a shows that the XGBoost models perform well based on Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE). Barisal division has the lowest error rates according to the MAE and RMSE, while the highest error rates are in Dhaka division. The MAE in Dhaka division is 109.93 and the RMSE is 127.39, which is likely due to unanticipated high-level dengue cases which impact forecast accuracy. According to MAPE, Dhaka has the highest ability to forecast compared to all other cities with 12.9%.The model predicts a decline in Dhaka's values from about 279 on December 1, 2023, to around 104 by December 14, 2023. In contrast, Chittagong is expected to remain stable at around 237, Barisal at approximately 125, and Khulna will fluctuate between 141 and 143 with a slight upward trend. Other divisions are projected to stay stable at about 187. Dhaka shows the most significant variation and upward trend compared to the other divisions, which are more stable or show minor fluctuations.

Table 5b summarizes the predictive accuracy of the best models for forecasting dengue cases in five divisions of Bangladesh, using mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE). Dhaka shows the highest MAE (820.30) and RMSE (850.76), indicating significant variability in dengue cases, while its MAPE (0.857) is comparable to Chattogram's (0.752). In contrast, Barisal, Khulna, and Other Divisions have much lower MAE and RMSE, with Other Divisions recording the lowest RMSE (66.12) and MAE (65.92). However, MAPE values are high in Barisal (0.731), Khulna (0.718), and Other Divisions (0.685), indicating greater proportional errors. Overall, while Khulna and Other Divisions show the best absolute accuracy, Dhaka and Chattogram excel in proportional accuracy, highlighting regional differences in model performance due to varying case counts and outbreak dynamics. The analysis confirms the model's adequacy, with Khulna and Other Divisions having the lowest error rates, and Dhaka the highest. The model predicts a decrease in dengue cases in Dhaka from about 240 on December 1, 2023, to 104 by December 14, 2023.

Table 5c shows the predictive accuracy of the best models for forecasting dengue cases in five divisions of Bangladesh, assessed using mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE). The MLP model shows its highest absolute errors in Dhaka (MAE = 850.215, RMSE = 940.225) but achieves the lowest relative error there (MAPE = 72.6%), whereas Chattogram records moderate absolute errors (MAE = 263.347, RMSE = 388.107) with a slightly higher MAPE of 77.7%. Barisal and Khulna attain the smallest absolute errors (MAE≈237 and 216; RMSE≈257 and 240, respectively) yet exhibit very high proportional errors (MAPE ≈91%), and the aggregated "Other Divisions" similarly yield low absolute errors (MAE = 229.644, RMSE = 293.851) alongside the highest relative error (MAPE = 91.7%), underscoring that in lower-incidence regions even modest absolute deviations translate into large percentage discrepancies. Overall, while Khulna and Barisal achieve high absolute accuracy, Dhaka offers the best proportional accuracy, highlighting regional differences in model performance due to variations in case volume and outbreak dynamics. The model predicts a gradual increase in Dhaka's values from about 340 on December 1, 2023, to around 358 by December 14, 2023. In contrast, Chattogram's values are expected to fluctuate between 74 and 79, Barisal's forecast remains stable at 23–24 until December 14, and Khulna's values are projected to vary between 22 and 23, showing a slight upward trend. Other divisions start at approximately 23 on December 1 and are expected to rise to 25 by December 14. Dhaka shows the most significant variation and upward trend compared to the other divisions, which remain relatively stable or show minor fluctuations.

Discussion

Temperature is crucial in predictive models, particularly for climate and weather forecasting. Our analysis indicates that it is the top predictor of future trends, with an importance value of about 0.6. While humidity and rainfall are also important, their influence is comparatively lower due to complex interactions and indirect effects. Focusing on temperature in data collection and feature selection can significantly improve model

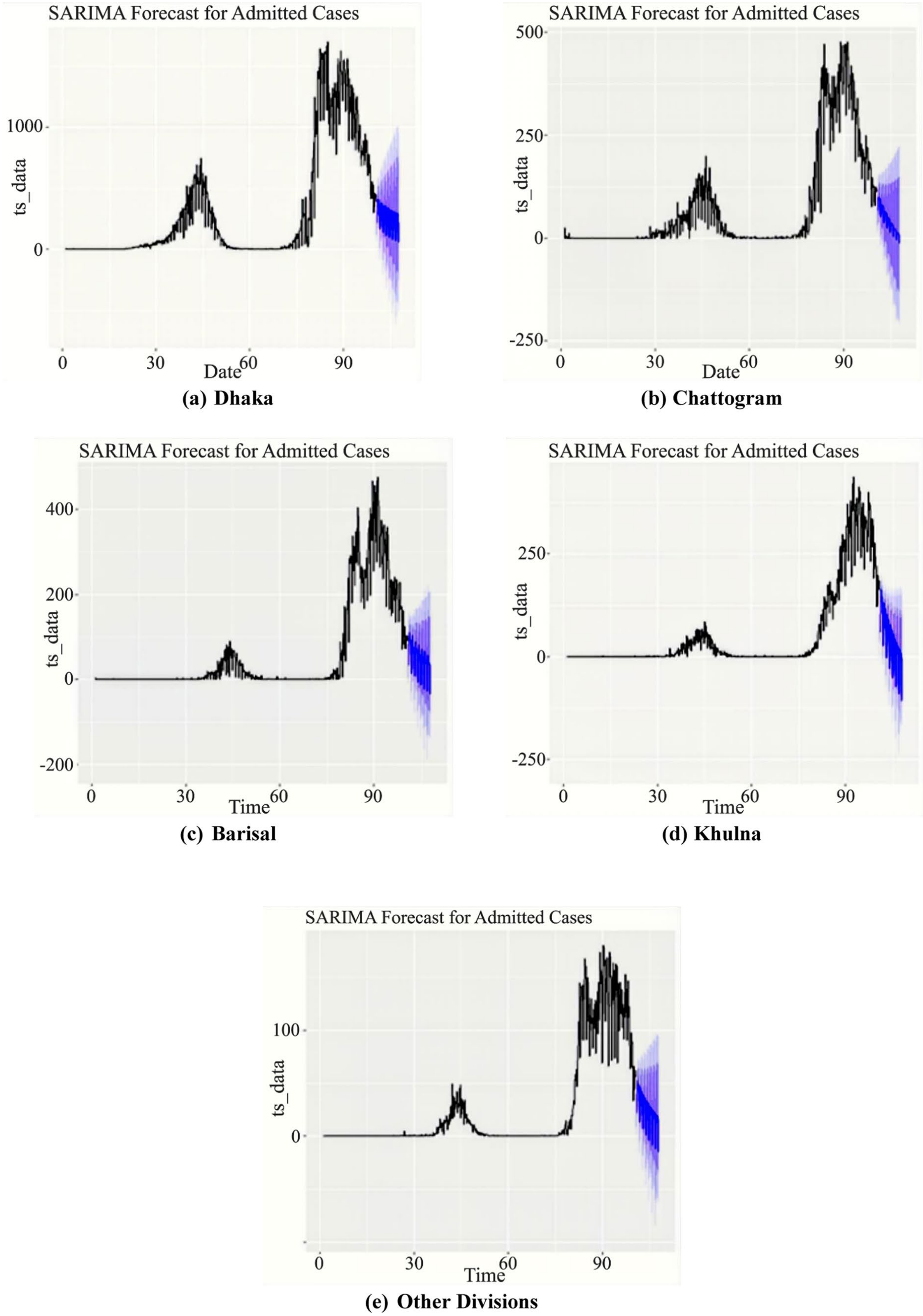
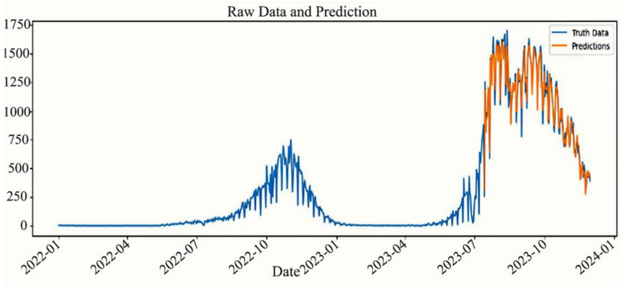
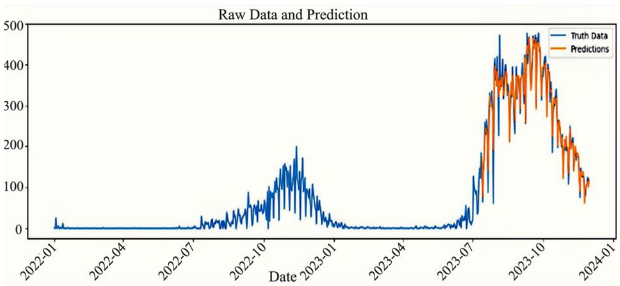


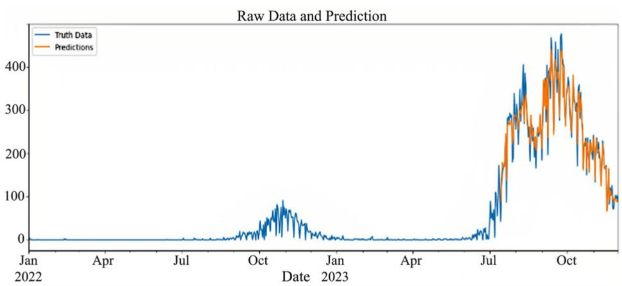
Fig. 4. SARIMA Forecast Plot of No. of Cases for the Divisions of Bangladesh.



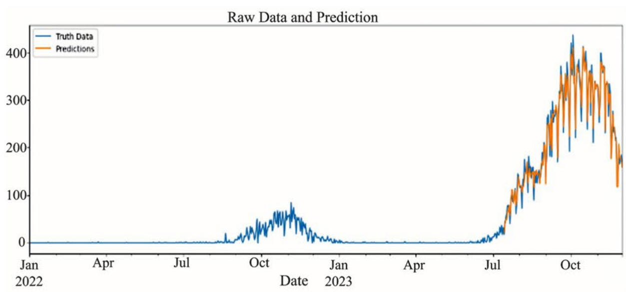
(a) Dhaka



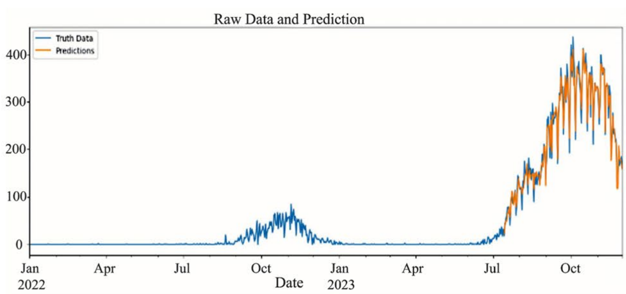
(b) Chattogram



(c) Barisal

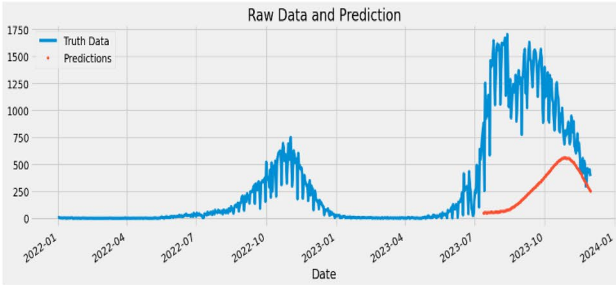


(d) Khulna

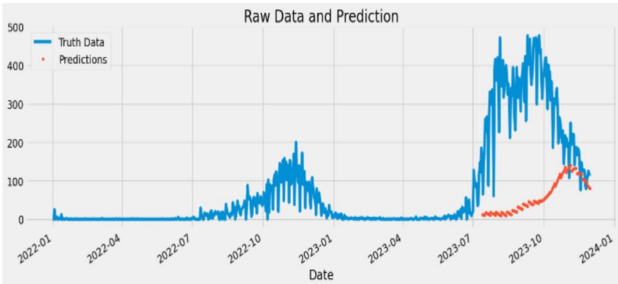


(e) Other Divisions

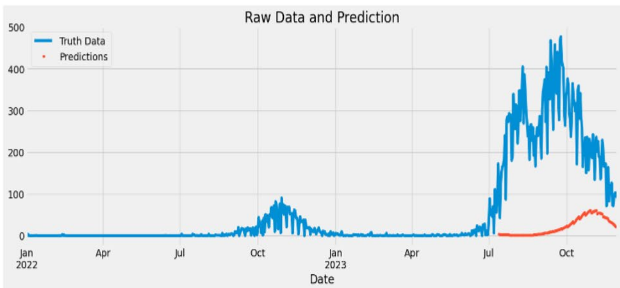
Fig. 5. Predicted plot for the no. of Dengue Cases among the Divisions of Bangladesh using XGBoost model.



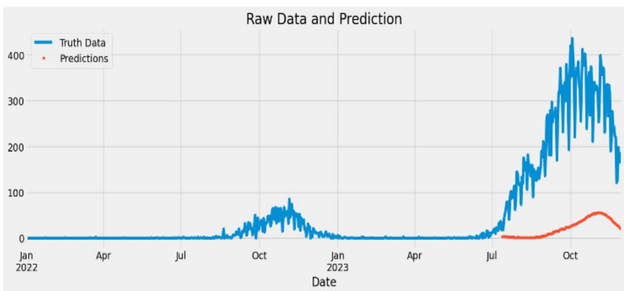
(a) Dhaka



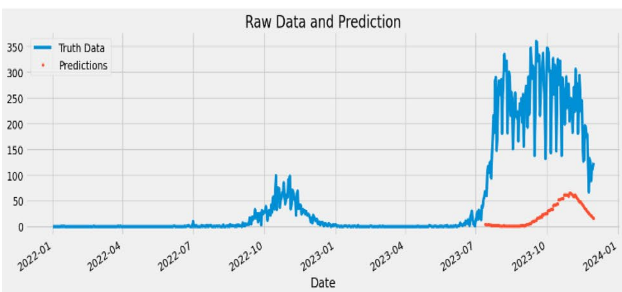
(b) Chattogram



(c) Barisal

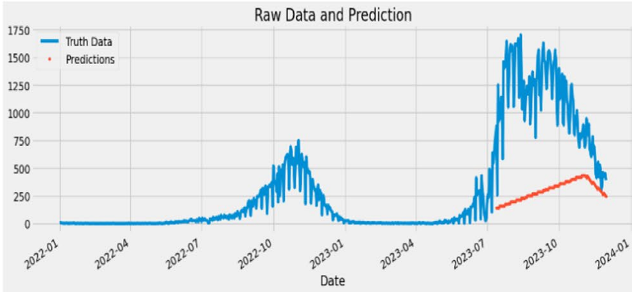


(d) Khulna

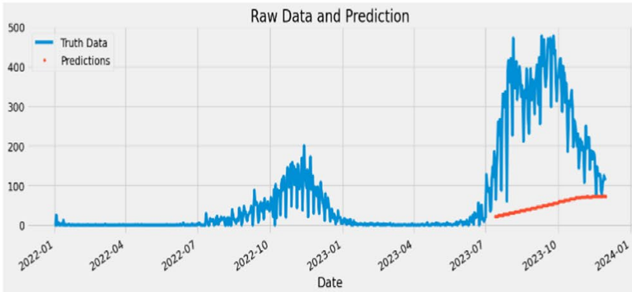


(e) Other Divisions

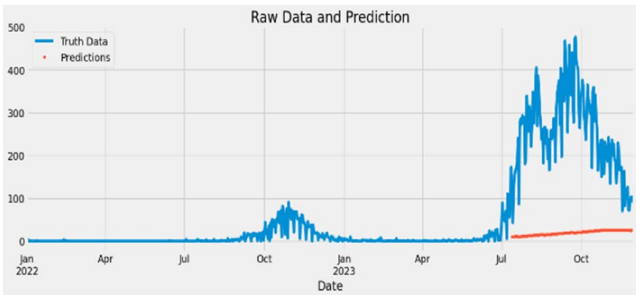
Fig. 6. Predicted plot for the no. of Dengue Cases among the Divisions of Bangladesh using Support Vector Regression (SVR) model.



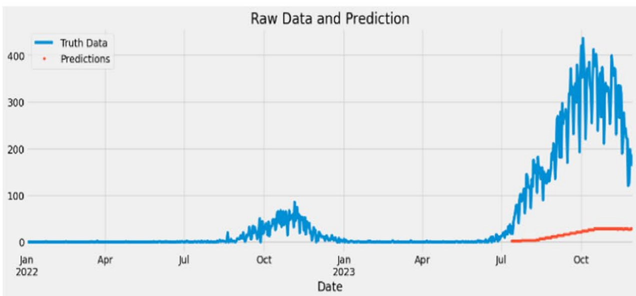
(a) Dhaka



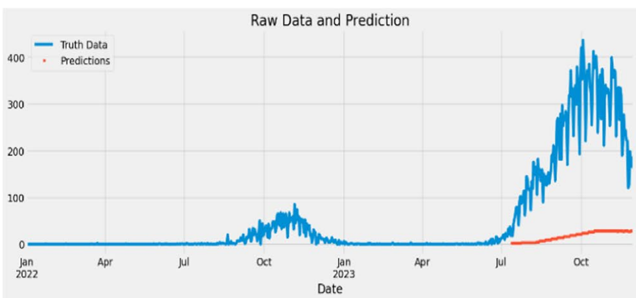
(b) Chattogram



(c) Barisal



(d) Khulna



(e) Other Divisions

Fig. 7. Predicted plot for the number of Dengue Cases among the Divisions of Bangladesh using multilayer perceptron (MLP).

Model	Divisions	MAE	RMSE	MAPE
(a) XGBoost	Dhaka	109.93	127.39	0.129
	Chattogram	103.65	110.56	0.469
	Barisal	31.00	31.87	0.269
	Khulna	41.42	41.94	0.434
	Other Divisions	41.06	41.39	0.425
(b) SVR	Dhaka	820.30	850.76	0.857
	Chattogram	165.11	172.14	0.752
	Barisal	83.53	83.97	0.731
	Khulna	68.65	69.32	0.718
	Other Divisions	65.92	66.12	0.685
(c) MLP	Dhaka	850.215	940.225	0.726
	Chattogram	263.347	388.107	0.777
	Barisal	237.441	256.947	0.914
	Khulna	216.192	239.774	0.911
	Other Divisions	229.644	293.851	0.917

Table 5. Best Model States for No. of Dengue Cases of four divisions using XGBoost.

performance and accuracy, as highlighted by research on its importance for disaster management, agriculture, and water resources. The use of a SARIMA model for predicting dengue cases in Bangladesh demonstrates its effectiveness in analyzing time series data. Our findings reveal that the model identifies significant patterns and seasonal changes, performing well in data-rich areas like Dhaka, but showing variability in regions like Khulna with limited data. This variability emphasizes the need for better data collection. The model's capacity to detect seasonal shifts is vital for public health planning, supported by numerous studies on time series analysis and epidemiological forecasting.

Alongside the SARIMA model, we used various machine learning techniques, including XGBoost, Support Vector Regression (SVR), and Multi-Layer Perceptron (MLP). XGBoost, known for its strong gradient boosting capabilities, was applied to analyze dengue incidence data. Its effectiveness in predicting outbreaks in different areas of Bangladesh highlights the importance of data quality and availability. In data-rich regions like Dhaka, the model accurately captures seasonal trends, enabling precise predictions and timely public health responses. Conversely, in data-scarce areas like Khulna, the model's performance varies significantly, indicating a need for improved data collection to enhance predictive accuracy. This aligns with numerous studies emphasizing the critical role of high-quality data in public health planning[42].

Our analysis indicates that Dhaka surpasses all other models, primarily because of its environmental conditions. The region experiences elevated temperatures that promote the transmission of dengue. Furthermore, the number of patients hospitalized in Dhaka is greater than in other divisions.

The evaluation of SARIMA and XGBoost models for predicting dengue outbreaks in Bangladesh highlights the strengths of each method. The SARIMA model, designed for time series analysis, shows higher accuracy in Dhaka and Barisal, evidenced by lower Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). However, it has a higher Mean Absolute Percentage Error (MAPE) than XGBoost, which affects its forecasting precision. In contrast, XGBBOST is a powerful deep learning technique that excels with large and complex datasets. Our results show that XGBoost surpasses traditional SARIMA models in predictive performance, particularly in data-rich areas like Dhaka, stressing the need for thorough data collection. While XGBoost is adaptable across various predictive contexts, it also highlights the critical role of data quality in ensuring reliable forecasts for public health initiatives. This analysis suggests that advanced machine learning methods like XGBoost may offer improved predictive capabilities for dengue outbreak forecasting compared to conventional models like SARIMA[43].

Although XGBoost performs better than other models according to model assessment metrics (RMSE, MAE, and MAPE), this approach's true worth is in its ability to explain. Temperature (importance = 0.6) is the primary driver of dengue incidence, followed by humidity (0.3) and rainfall (0.1), according to the feature importance analysis (Fig. 2, Table 2). Epidemiological data suggesting warmer climates speed up virus replication and mosquito reproduction is in line with this finding.

Forecasts tailored to a particular division also offer practical solutions. Interventions like focused vector control during the hottest summer months and increased hospital readiness are essential in Dhaka, where high temperatures and a dense population mix. The combined effects of rainfall and humidity in Chattogram highlight the significance of pre-monsoon awareness efforts and drainage upgrades. Even though the incidence is lower in Barisal, Khulna, and other divisions, the model emphasizes the necessity of proactive surveillance systems to stop localized outbreaks from getting worse.

As a result, XGBoost offers a mechanistic explanation of dengue dynamics in addition to greater predicting accuracy. Policymakers can better allocate resources to reduce dengue transmission across divisions by associating feature importance with customized treatments.

Conclusion

The comparative analysis of various machine learning algorithms for predicting dengue incidence in Bangladesh reveals the strengths and weaknesses of different modeling techniques. Traditional time series models, such as SARIMA, excel at detecting seasonal trends and patterns; however, they tend to have a higher Mean Absolute Percentage Error (MAPE) than XGBoost, leading to reduced forecasting accuracy. Machine learning and deep learning methods play a crucial role in shaping public health strategies, particularly due to their capacity to manage the complex, non-linear relationships found in epidemiological data. Advanced machine learning models like XGBoost demonstrate significant robustness in processing intricate datasets and can adapt effectively to varying data conditions. Our research supports this observation, as indicated by the lower MAPE values associated with XGBoost compared to traditional SARIMA models.

The study's findings emphasize the critical role of high-quality, comprehensive data in enhancing the predictive accuracy of these models. By leveraging the complementary strengths of both SARIMA and XGBoost, a hybrid approach can be developed that maximizes predictive performance and provides more reliable forecasts for public health interventions. This integrative strategy can significantly improve dengue outbreak management and resource allocation in Bangladesh, supporting more effective and timely public health responses.

Data availability

The datasets analysed during the current study are available from the corresponding author ([mxh220100@utdallas.edu] (mailto:mxh220100@utdallas.edu)) on reasonable request.

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Declarations

Competing interests
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